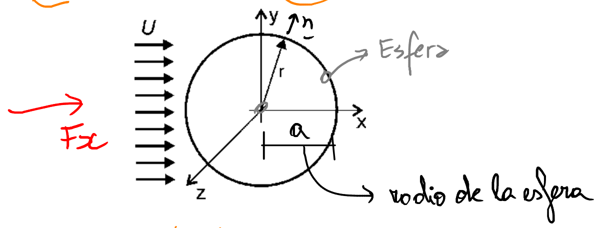
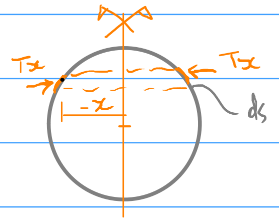
$$\underline{f} = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} \qquad \underline{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \qquad \underline{T} = \underline{\Upsilon} \cdot \underline{n}$$

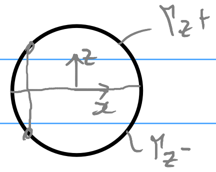
$\underline{n}$ → normal a la esfera

$$T_x = -\frac{x}{a} p_0 + \frac{3}{2}\mu \frac{U}{a}; \quad T_y = -\frac{y}{a} p_0; \quad T_z = -\frac{z}{a} p_0.$$

$$\underline{F} = \int_\Gamma \underline{T}\, ds = \begin{bmatrix} \int_\Gamma T_x\, ds \\ \int_\Gamma T_y\, ds \\ \int_\Gamma T_z\, ds \end{bmatrix}$$







$$F_z = \int_\Gamma T_z\, ds = -\frac{p_0}{a}\int_\Gamma z\, ds = -\frac{p_0}{a}\left(\cancel{\int_{\Gamma_{z+}} z\, ds} + \cancel{\int_{\Gamma_{z-}} z\, ds}\right) = 0$$

$$F_y = \int_\Gamma T_y\, ds = -\frac{p_0}{a}\int_\Gamma y\, ds = -\frac{p_0}{a} \cdot 0 = 0$$

$$F_x = \int_\Gamma T_x\, ds = \int_\Gamma \left(\cancel{-\frac{x}{a}}^{\,0} p_0 + \frac{3}{2}\mu\frac{U}{a}\right) ds = \frac{3}{2}\mu\frac{U}{a}\underbrace{\int_\Gamma 1\, ds}_{\text{superficie de la esfera}}$$

$$= \frac{3}{\cancel{2}_{1}}\mu\frac{U}{\cancel{a}}\,\cancel{4}^{2}\pi a^{\cancel{2}} = 6\mu U \pi a = \underline{6\pi a \mu U}$$